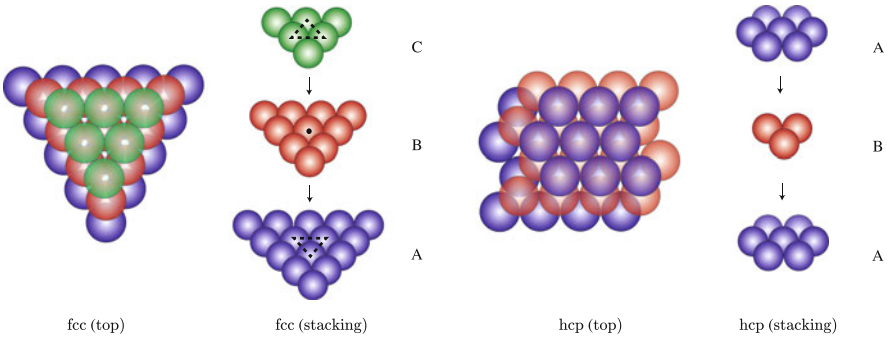


Fig. 1.18 The face centered cubic and hexagonally close packed lattice are associated to two different stacking arrangements of hexagonally structured planar lattices. The *dashed triangles* indicated in the upper and lower layers of the fcc lattice are centered on the atom indicated by a *dot* in the middle layer

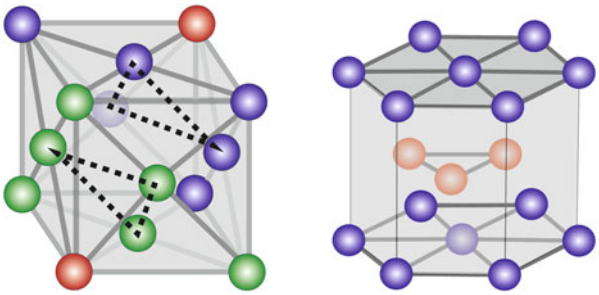


Fig. 1.19 Unit cells for fcc (*center*), and hcp (*right*) geometries. The coloring of atoms and *dashed triangles* are indicated to match the stacking of layers in Fig. 1.18

(hcp) lattices. The fcc lattice corresponds to the common arrangement by which cannonballs are stacked into pyramidal structures; it can be viewed as a periodic stacking (ABCABC...) of three hexagonally structured planar layers, as illustrated in Fig. 1.18. Also shown in Fig. 1.18 is the hcp lattice, which, on the other hand, alternates two distinct planar lattices. Layer stackings then define a certain unit cell as shown in Fig. 1.19.

1.6.1 Lattice Vibrations

Regardless of the choice of boundary and/or the inclusion of non-pairwise potentials, the minimum of the potential energy occurs where

$$\nabla U = \mathbf{0},$$

which gives in general a nonlinear system of N_c equations in N_c unknowns to be solved for the position vector $\boldsymbol{q}^*$ associated to mechanical equilibrium.

At the equilibrium point, we can linearize the system of differential equations by computing the Hessian matrix, then we find

$$\nabla U(\boldsymbol{q}) \approx U''(\boldsymbol{q}^*)(\boldsymbol{q} - \boldsymbol{q}^*).$$

Then, letting $\delta\boldsymbol{q} = \boldsymbol{q} - \boldsymbol{q}^*$, $\delta\boldsymbol{p}$ represent small deviations from the equilibrium point at $(\boldsymbol{q}, \boldsymbol{p}) = (\boldsymbol{q}^*, \boldsymbol{0})$, we have

$$\frac{\mathrm{d}\delta\boldsymbol{q}}{\mathrm{d}t} = \boldsymbol{M}^{-1}\delta\boldsymbol{p},$$
$$\frac{\mathrm{d}\delta\boldsymbol{p}}{\mathrm{d}t} = -U''(\boldsymbol{q}^*)\delta\boldsymbol{q}.$$

At the minimum of the potential energy, U'' is a positive definite symmetric matrix. The eigenvalues of the matrix

$$\boldsymbol{A} := \begin{bmatrix} \boldsymbol{0} & \boldsymbol{M}^{-1} \\ -U''(\boldsymbol{q}^*) & \boldsymbol{0} \end{bmatrix}$$

are therefore all purely imaginary ($\pm i\Omega$, $\Omega \in \mathbb{R}$). Associated to each eigenvalue pair we have a pair of complex conjugate eigenvectors $\boldsymbol{\xi}, \bar{\boldsymbol{\xi}}$ and also a pair of solutions which can be written in the complex form

$$z(t) = ae^{i\Omega t}\boldsymbol{\xi} + be^{-i\Omega t}\bar{\boldsymbol{\xi}},$$

(a, b complex coefficients), or recast in real form as (α, β real coefficients):

$$z(t) = \alpha[\sin(\Omega t)\mathrm{Re}(\boldsymbol{\xi}) + \cos(\Omega t)\mathrm{Im}(\boldsymbol{\xi})] + \beta[\cos(\Omega t)\mathrm{Re}(\boldsymbol{\xi}) - \sin(\Omega t)\mathrm{Im}(\boldsymbol{\xi})].$$

These are referred to as the "normal modes" of oscillation of the system (Fig. 1.20 illustrates the normal modes for a linear molecule).

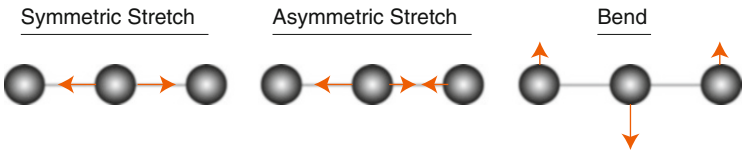


Fig. 1.20 Three normal modes of a linear molecule (i.e., a molecule in which the atoms, at minimum energy, are arranged on a line)